

Disproof of conjecture 00000008417

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Verdict: FALSE

We disprove conjecture 00000008417 as stated. The witness is the small Witt design $W_{12} = S(5, 6, 12)$: a genuine 5-(12, 6, 1) design with 132 blocks. It exists at $n = 12$, and at $(t, k, n) = (5, 6, 12)$ all six classical divisibility conditions hold. Hence the existence threshold obeys $n_0(5, 6) \leq 12$, which contradicts the filed conjunct “the lower bound $n_0(5, 6) > 12$ is given by a concrete divisibility obstruction”. In fact there is no divisibility obstruction at $n = 12$ at all. The refutation is unconditional and completely elementary; it does not rely on any unproved hypothesis.

1 The conjecture, quoted verbatim

From `conjectures/00000008417.md`:

English. Definition: A Steiner system $S(t, k, n)$ is a design in which every t -element subset lies in exactly one block. Conjecture: $S(t, k, n)$ exists if and only if the classical divisibility conditions hold and n is at least $n_0(t, k)$; $n_0(t, k)$ admits an explicit bound of type $(kt)^{ct}$, and the lower bound $n_0(5, 6) > 12$ is given by a concrete divisibility obstruction. (explicit threshold for Steiner systems)

The conjecture is filed in both English and Chinese; the Chinese original is reproduced verbatim (in Unicode) in `README.md`. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

We read the conjecture as the conjunction of two clauses:

- (C1) (*the existence criterion*) for every (t, k, n) with $t < k \leq n$, $S(t, k, n)$ exists if and only if the classical divisibility conditions hold and $n \geq n_0(t, k)$;
- (C2) (*the small- n lower bound*) $n_0(5, 6) > 12$, and this bound is “given by a concrete divisibility obstruction”.

We show that (C1) and (C2) cannot both be true, and that (C2) is false as filed.

2 The divisibility conditions at $(5, 6, 12)$

For an $S(t, k, n)$ the classical divisibility conditions are

$$\lambda_i := \frac{\binom{n-i}{t-i}}{\binom{k-i}{t-i}} \in \mathbb{Z}, \quad i = 0, 1, \dots, t,$$

where λ_i is the number of blocks through a fixed i -subset (λ_0 is the total number of blocks). At $(t, k, n) = (5, 6, 12)$:

$$\lambda_0 = \frac{\binom{12}{5}}{\binom{6}{5}} = \frac{792}{6} = 132, \quad \lambda_1 = \frac{\binom{11}{4}}{\binom{5}{4}} = \frac{330}{5} = 66, \quad \lambda_2 = \frac{\binom{10}{3}}{\binom{4}{3}} = \frac{120}{4} = 30,$$

$$\lambda_3 = \frac{\binom{9}{2}}{\binom{3}{2}} = \frac{36}{3} = 12, \quad \lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = \frac{8}{2} = 4, \quad \lambda_5 = \frac{\binom{7}{0}}{\binom{1}{0}} = \frac{1}{1} = 1.$$

Every quotient is an integer. We record this as exact products:

i	exact identity	quotient λ_i
0	$792 = 132 \cdot 6$	132
1	$330 = 66 \cdot 5$	66
2	$120 = 30 \cdot 4$	30
3	$36 = 12 \cdot 3$	12
4	$8 = 4 \cdot 2$	4
5	$1 = 1 \cdot 1$	1

Remark 1. Thus the six conditions hold simultaneously at $n = 12$. There is *no* divisibility obstruction at $n = 12$, in direct contradiction to the claim that the bound $n_0(5, 6) > 12$ is supplied by such an obstruction.

3 The existence certificate

Definition 2. A t -($n, k, 1$) design (Steiner system $S(t, k, n)$) is a family $\mathcal{B}$ of k -subsets (blocks) of an n -set such that every t -subset lies in exactly one block. For $t = 5$, $k = 6$, $n = 12$ the number of blocks must be $\lambda_0 = \binom{12}{5} / \binom{6}{5} = 132$.

Theorem 3. *There exists a 5-(12, 6, 1) design: 132 six-element blocks on the point set $\{0, 1, \dots, 11\}$ such that every one of the $\binom{12}{5} = 792$ five-element subsets is contained in exactly one block.*

Two independent constructions, each exhaustively verified. We give two constructions and check both against all 792 five-subsets.

(a) *Exact cover / backtracking.* Let the columns be the 792 five-subsets and the rows the $924 = \binom{12}{6}$ six-subsets; selecting a row covers its $\binom{6}{5} = 6$ five-subsets. An Algorithm X backtracking search with the minimum-remaining-values heuristic over the 792 columns selects 132 rows. Direct enumeration confirms that all 792 five-subsets are covered *exactly once* (each count equals 1), that the 132 blocks are distinct and all have size 6. This is the design whose 132 12-bit masks appear verbatim in `lean4/Main.lean`.

(b) *Extended ternary Golay code.* The extended ternary Golay code $[12, 6, 6]$ over $\mathbb{F}_3$ has weight distribution

$$A_0 = 1, \quad A_6 = 264, \quad A_9 = 440, \quad A_{12} = 24$$

and $3^6 = 729$ codewords. Its 264 weight-6 words split into 132 opposite pairs $\{\pm c\}$, giving 132 distinct 6-element supports. Exhaustive enumeration of all 792 five-subsets confirms that these 132 supports also cover every five-subset exactly once.

The two designs are isomorphic. An explicit point permutation $p = (0, 1, 2, 3, 4, 11, 7, 9, 8, 6, 5, 10)$ maps the exact-cover design onto the Golay design; i.e. the two labelled designs coincide up to relabelling. The common design is W_{12} . $\square$

Remark 4 (Invariants and uniqueness). Both constructions satisfy the same combinatorial invariants: the $132 \cdot 131/2 = 8646$ pairs of blocks intersect in $0, 2, 3, 4$ points with multiplicities $66, 2970, 2640, 2970$ respectively (intersections of size 1 or 5 never occur), and each block meets the others in i points with multiplicities $1, 0, 45, 40, 45, 0, 1$ for $i = 0, \dots, 6$. The design is classically unique up to isomorphism, with automorphism group M_{12} of order 95040; `reproduce.py` confirms this computationally by enumerating all 95040 automorphisms of the constructed design.

4 The logical dichotomy

The conjecture is a conjunction, and no reading makes both clauses true.

Theorem 5. *Let $n_0(5, 6)$ be the threshold appearing in clause (C1). Then:*

- (i) *if (C1) holds, then $n_0(5, 6) \leq 12$, so (C2) is false;*
- (ii) *if (C2) holds, then (C1) is false.*

Hence $(C1) \wedge (C2)$ is false under every reading.

Proof. By Theorem 3 there is an $S(5, 6, 12)$, and by Section 2 the six divisibility conditions hold at $n = 12$.

(i) Assume (C1). Instantiating it at $(t, k, n) = (5, 6, 12)$ gives

$$(S(5, 6, 12) \text{ exists}) \iff (\text{divisibility holds at } (5, 6, 12) \wedge 12 \geq n_0(5, 6)).$$

The left side is true (Theorem 3) and the divisibility conjunct on the right is true (Section 2), so the equivalence forces $12 \geq n_0(5, 6)$, i.e. $n_0(5, 6) \leq 12$. The literal statement of (C2), $n_0(5, 6) > 12$, is therefore false, and so is the claim that this bound is “given by a concrete divisibility obstruction” (there is no obstruction at $n = 12$: all six conditions hold).

(ii) Assume (C2), i.e. $n_0(5, 6) > 12$. At $(t, k, n) = (5, 6, 12)$ the divisibility conditions hold (Section 2) but $12 \geq n_0(5, 6)$ fails, so the right side of (C1) is false, while the left side is true by Theorem 3. Thus the asserted “if and only if” in (C1) is false.

Either way the conjunction $(C1) \wedge (C2)$ fails. This holds whether $n_0(t, k)$ is read as the exact least admissible n , as any threshold function making (C1) true, or as the explicit bound of type $(kt)^{ct}$ mentioned in the conjecture: any such quantity is forced by (C1) to satisfy $n_0(5, 6) \leq 12$. $\square$

Corollary 6. *Conjecture 00000008417 is FALSE.*

Proof. The conjecture asserts $(C1) \wedge (C2)$, which Theorem 5 shows to be unsatisfiable. Clause (C2) is independently and concretely false: the divisibility conditions hold at $n = 12$ and an $S(5, 6, 12)$ exists, so the threshold is at most 12. $\square$

5 Threshold context

In every small case the first admissible proper value of n is attained, and the extremal examples are exactly the Witt designs.

(t, k)	first admissible n	design	name
$(2, 3)$	7	$S(2, 3, 7) = \text{PG}(2, 2)$	Fano plane
$(3, 4)$	8	$S(3, 4, 8) = \text{AG}(3, 2)$	affine geometry
$(4, 5)$	11	$S(4, 5, 11)$	Witt design W_{11}
$(5, 6)$	12	$S(5, 6, 12)$	Witt design W_{12}
$(5, 8)$	24	$S(5, 8, 24)$	Witt design W_{24}

The $(5, 6)$ row is decisive: divisibility holds at $n = 12$ and the design exists there, so the threshold satisfies $n_0(5, 6) \leq 12 < n_0(5, 6)$ under the filed bound.

6 Scope note

We are explicit about what is and is not in dispute.

- The general large- n criterion “divisibility implies existence for $n \geq n_0(t, k)$ ” is essentially Keevash’s theorem. That theorem is non-effective and there is no explicit $(kt)^{ct}$ threshold in the literature. *This submission does not dispute the large- n existence direction.* It targets the definite, false conjunct $n_0(5, 6) > 12$ (and the accompanying “concrete divisibility obstruction”).
- At $n = 12$ the divisibility conditions hold and the design exists; hence no lower bound above 12 can be correct for $(5, 6)$, and no divisibility obstruction at $n = 12$ exists. This is a finite, verified, unconditional refutation.
- Uniqueness of W_{12} up to isomorphism and its automorphism group M_{12} (order 95040) are classical; they are not needed for the refutation, which needs only one design. They are reported for context and checked computationally.

7 Formalisation

The existence certificate is formalised in `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). Points are encoded as 12-bit `Nat` masks. The central theorem is

```
theorem every_five_in_exactly_one_block :
  fives.all (fun m => covered m == 1) = true
```

proved by `decide`; it scans all 792 five-subsets against the 132 blocks of `blocks` and checks that each is contained in exactly one. The six divisibility identities (`div_i0`, ..., `div_i5`), `witt_is_design` (`IsDesign5612 blocks`) and `exists_S5612` ($\exists B$, B is a 5 -($12, 6, 1$) design) are also proved. The audit lake `env lean Check.lean` reports only the axiom `propext` — no `sorryAx`, and no `Lean.ofReduceBool` because the fully kernel-checked `decide` variant is used. See `lean4/README.md` for the build time (≈ 63 s) and the full statement table.

8 Reproducibility

`reproduce.py` uses the Python 3 standard library only and performs: (a) the six divisibility quotients, checked to be integers; (b) the exact-cover construction with an exhaustive check that every one of the 792 five-subsets lies in exactly one of the 132 blocks, plus a SHA-256 of the canonical block list; (c) the independent extended ternary Golay construction with the same exhaustive coverage check; (d) an explicit isomorphism between the two labelled designs.

```
$ python3 reproduce.py
...
PASS: all checks verified; conjecture 00000008417 is FALSE.
```

It prints the block count, the canonical SHA-256 (`6b2ab5e7...eb567`), the invariants, an explicit relabelling isomorphism, and $|\text{Aut}| = 95040$, and exits 0 exactly when every check holds.

```
$ tectonic --outdir build main.tex
$ ls -l build/main.pdf
```

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `Main`, `name = "tlmc8417"`, no dependencies), `Main.lean`, `Check.lean` (`#print axioms`), and `lean4/README.md`. `lake build` exits 0 and the audit reports no `sorryAx`.